

Local Structure Feature Kernel Specification

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Scope

This specification defines the analysis-owned per-atom local-structure feature kernel introduced for MLFF-DATA9A7b. It owns geometry, normalization, numerical feature definitions, missing-value semantics, warnings, and complexity failure. It does not own MLFF role authorization, atom-group aggregation, target-order metric fitting, selection quotas, or checkpoint decisions.

The equations below make the already implemented numerical contract reconstructible without reverse-engineering `mdstats.analysis.local_structure`. They are a documentation reconciliation of the existing feature-value method, not a change to that method.

API

```
LocalStructureFeaturePolicy
LocalStructureFeatureResult
compute_local_structure_features(
    collection,
    *,
    frame_index,
    atom_indices=None,
    policy=None,
)
```

`atom_indices` selects center atoms. Every atom in the collection remains eligible as a neighbor. The result contains one immutable row per selected center, a stable feature-name order, a finite binary64 value matrix, an aligned missing mask, warning codes, and policy metadata.

Geometry and pair domain

Coordinates use the collection cell and origin. Pair vectors use the shared triclinic minimum-image geometry and the collection periodic-boundary-condition (PBC) flags. The center atom is excluded from its own neighbor population. Distinct atoms at minimum-image distance less than or equal to `coincident_tolerance_angstrom` produce warning evidence. They remain valid for scalar distance, connectivity, radial, density, and species-mixing features, but are excluded from angular/orientational unit-vector features so division by a near-zero norm cannot occur.

The output scalar features must be invariant under:

- rigid translation;
- global rotation of positions and cell;
- atom permutation, after rows are mapped back to the same centers; and
- equivalent periodic images.

Covalent radii and smooth connectivity

For center i and distinct neighbor j , let r_{ij} be the minimum-image Cartesian distance. Let R_i and R_j be ASE covalent radii. If an atomic number has no finite positive ASE radius, use `fallback_covalent_radius_angstrom` and emit the corresponding fallback warning. Define

$$x_{ij} = \frac{r_{ij}}{R_i + R_j}.$$

With $a = \text{normalized_switch_start}$ and $b = \text{normalized_switch_end}$, the exact smooth connectivity weight is

$$w_{ij} = \begin{cases} 1, & x_{ij} \leq a, \\ \frac{1}{2} \left[1 + \cos\left(\pi \frac{x_{ij}-a}{b-a}\right) \right], & a < x_{ij} < b, \\ 0, & x_{ij} \geq b. \end{cases}$$

Let

$$W_i = \sum_j w_{ij}.$$

The scalar connectivity/distance features are:

$$d_i^{\min} = \min_j r_{ij},$$

$$\mu_{r,i} = \frac{\sum_j w_{ij} r_{ij}}{W_i},$$

$$\sigma_{r,i} = \sqrt{\frac{\sum_j w_{ij} (r_{ij} - \mu_{r,i})^2}{W_i}},$$

$$C_i = \sum_j w_{ij},$$

$$H_i = \#\{j : w_{ij} > w_{\min}\},$$

and

$$D_i = \sqrt{\sum_j w_{ij}^2},$$

where $w_{\min} = \text{minimum_weight}$.

If there is no distinct neighbor, `nearest_neighbor_distance_angstrom` is zero-filled and marked missing. If $W_i=0$, the weighted mean and weighted standard deviation are zero-filled and marked missing. `smooth_coordination`, `hard_neighbor_count`, and `weighted_degree_l2` remain defined as zero in that case.

These descriptors express smooth local support; they are not a universal chemical-bond definition.

Neighbor-species entropy

For species s , define its smooth neighbor mass

$$W_{i,s} = \sum_{j:Z_j=s} w_{ij}$$

and, when $W_i > 0$,

$$p_{i,s} = \frac{W_{i,s}}{W_i}.$$

The species-mixing feature is

$$S_i = - \sum_{s:p_{i,s}>0} p_{i,s} \log p_{i,s}.$$

If $w_i=0$, the stored value is zero and the feature is marked missing.

Local density and radial environment

Let $R_d = \text{density_radius_angstrom}$. The local number-density proxy is

$$\rho_i = \frac{\sum_j \exp[-(r_{ij}/R_d)^2]}{\frac{4}{3}\pi R_d^3}.$$

It is a Gaussian number-density proxy, not a Voronoi volume and not a claim of free volume.

For every configured radial center c and $\sigma_r = \text{radial_width_angstrom}$, define

$$G_i(c) = \sum_j \exp\left[-\frac{1}{2}\left(\frac{r_{ij}-c}{\sigma_r}\right)^2\right].$$

The radial coordinates are emitted in strictly increasing configured center order as `radial_density_r{center:.3f}_angstrom`.

Neither the density nor radial Gaussian uses the smooth connectivity weight; the minimum-image distance itself controls these kernels.

Angular Legendre moments

Angular/orientational neighbors are those satisfying both

$$w_{ij} > w_{\min}$$

and

$$r_{ij} > r_{\text{coincident}},$$

where $r_{\text{coincident}} = \text{coincident_tolerance_angstrom}$. For such a neighbor let

$$\hat{\mathbf{r}}_{ij} = \frac{\mathbf{r}_{ij}}{r_{ij}}.$$

For every configured Legendre order l , define the unordered-pair weight sum

$$W_i^{(2)} = \sum_{j<k} w_{ij} w_{ik}.$$

When at least two angular neighbors exist and $w_i^{(2)}>0$, the angular moment is

$$A_{i,l} = \frac{\sum_{j<k} w_{ij} w_{ik} P_l(\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{r}}_{ik})}{W_i^{(2)}},$$

where P_l is the Legendre polynomial of degree l . Otherwise the stored value is zero and the angular coordinate is marked missing.

Bond-orientational order

For every configured orientational order l , the weighted Steinhardt bond-orientational invariant is defined by the spherical-harmonic addition theorem. With the angular-neighbor set above and

$$W_i^{(1)} = \sum_j w_{ij},$$
$$q_{i,l}^2 = \frac{\sum_j w_{ij}^2 + 2 \sum_{j < k} w_{ij} w_{ik} P_l(\hat{\mathbf{r}}_{ij} \cdot \hat{\mathbf{r}}_{ik})}{(W_i^{(1)})^2},$$

and

$$q_{i,l} = \sqrt{\max(0, q_{i,l}^2)}.$$

This is equivalent to the conventional weighted spherical-harmonic definition

$$q_l = \sqrt{\frac{4\pi}{2l+1} \sum_{m=-l}^l |q_{lm}|^2}.$$

If no angular/orientational neighbor remains, the stored value is zero and the coordinate is marked missing. One valid neighbor is sufficient to define the orientational invariant; the two-neighbor requirement applies to the unordered pair angular moment, not to q_l .

Feature order and default numerical policy

The stable feature order is:

1. nearest_neighbor_distance_angstrom;
2. weighted_neighbor_distance_mean_angstrom;
3. weighted_neighbor_distance_std_angstrom;
4. smooth_coordination;
5. hard_neighbor_count;
6. weighted_degree_l2;
7. neighbor_species_entropy;
8. local_number_density_angstrom⁻³;
9. radial coordinates in configured center order;
10. angular_legendre_l{order} in increasing configured order; and
11. bond_orientational_q{order} in increasing configured order.

The current default policy is:

```
normalized_switch_start = 1.15
normalized_switch_end = 1.75
radial_centers_angstrom = (1.0,1.5,2.0,2.5,3.0,3.5,4.0,5.0)
radial_width_angstrom = 0.35
density_radius_angstrom = 4.0
angular_legendre_orders = (1,2,3,4)
orientational_orders = (4,6)
minimum_weight = 1e-8
coincident_tolerance_angstrom = 1e-8
fallback_covalent_radius_angstrom = 1.0
maximum_dense_pair_work = 4_000_000
policy_version = mdstats.analysis.local-structure.2026-07.v1
```

Policy values must be finite and positive where applicable; switch start must be less than switch end; radial centers are strictly increasing; and `minimum_weight < 1`.

Missing values and finite representation

Every returned numerical array is finite. Undefined coordinates are represented by value zero plus an aligned `missing_mask=true`; zero is therefore not by itself evidence that a quantity was observed. Current missing rules are:

- no distinct neighbor -> nearest-neighbor distance missing;
- zero smooth weight sum -> weighted distance mean/std and species entropy missing;
- fewer than two valid angular neighbors or zero angular pair-weight sum -> each angular Legendre moment missing;
- no valid orientational neighbor -> each bond-orientational invariant missing.

Other coordinates remain defined by the formulas above, including zero-valued coordination/count/degree, radial, and density results.

Precision and backend equivalence

Geometry and feature arrays are evaluated and returned in IEEE-754 binary64 under the current implementation. The mathematical formulas, feature order, missing-mask semantics, and policy values above are normative; a particular NumPy/SciPy reduction kernel, vectorization strategy, fused/non-fused execution choice, or loop decomposition is not independently promoted to analysis semantic identity unless an accepted numerical owner explicitly does so.

A replacement backend must preserve the public feature contract and pass backend-equivalence evidence appropriate to its consumers. A consumer whose later discrete decision can change under provider roundoff must additionally bind the provider policy/version and satisfy that consumer's precision/ discrete-decision equivalence contract rather than assuming arbitrary backend roundoff is harmless.

Complexity and backend

The current implementation evaluates a dense center-by-population pair matrix. Before allocation it evaluates

$$W = N_{\text{centers}} N_{\text{atoms}}.$$

If W exceeds `maximum_dense_pair_work`, the function raises `LocalStructureComplexityError`. No silent subsampling or policy relaxation is allowed. A future cell-list or Verlet implementation may replace the kernel only if it preserves the feature contract above and passes backend-equivalence tests.

Ownership and use

The MLFF branch may:

- call this API on authorized frames;
- aggregate rows by immutable atom groups or by an independently defined neutral element view;
- use the resulting descriptors for fitted metrics and coverage selection; and
- record generic structural changes.

The MLFF branch may not redefine the switching function, minimum-image semantics, radial/density kernels, angular normalization, orientational normalization, or missing-value rules.

The kernel is not automatically an `ObservableAnalysisCall`. Validation-grade radial distribution functions (RDFs), coordination distributions, angle distributions, connectivity, and future orientational-order distributions retain their own result schemas.

Acceptance tests

Required tests include:

1. rigid translation and rotation invariance;
2. atom-permutation equivalence;
3. orthorhombic and triclinic periodic minimum images;
4. exact switch fixtures below, inside, and above the transition interval;
5. direct scalar references for weighted mean/std, coordination, hard-neighbor count, weighted degree, and species entropy;
6. direct scalar references for radial and local-density Gaussian kernels;
7. direct scalar Legendre-moment and bond-orientational fixtures, including the one-neighbor orientational case;
8. explicit angular/orientational missing masks;
9. selected-center behavior;
10. policy round trip and stable feature order;
11. coincident/fallback warnings; and
12. fail-closed pair-work budget.